

AGENTIC AI-BASED MOTOR INSURANCE CLAIMS PROCESSING AND POLICY SUPPORT SYSTEM

Proposed Solution, Architecture, Implementation Plan, Responsible AI and Commercialization

Domain: Insurance - Motor Insurance Claims and Policy Support

Core principle: AI assists with understanding, retrieval, triage and guidance; human claims officers retain authority for final claim decisions.

Prepared for Mid Evaluation / Project Proposal

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Section	Purpose
1. Why We Selected the Domain	Problem, users, importance and suitability for Agentic AI
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4. Complete Workflows	Information-request and claim-submission flows
5. Implementation Plan	IR, NLP, orchestration, fraud triage, LLM, APIs and security
6. Responsible AI Plan	Fairness, transparency, explainability, privacy and human oversight
7. Commercialization Plan	Customers, value proposition, revenue model and rollout
8. Complete Proposed Solution	Concise presentation-ready explanation

1. Why We Selected the Domain

1.1 Domain

Selected domain: Insurance, specifically Motor Insurance Claims and Policy Support.

1.2 Real-World Problem

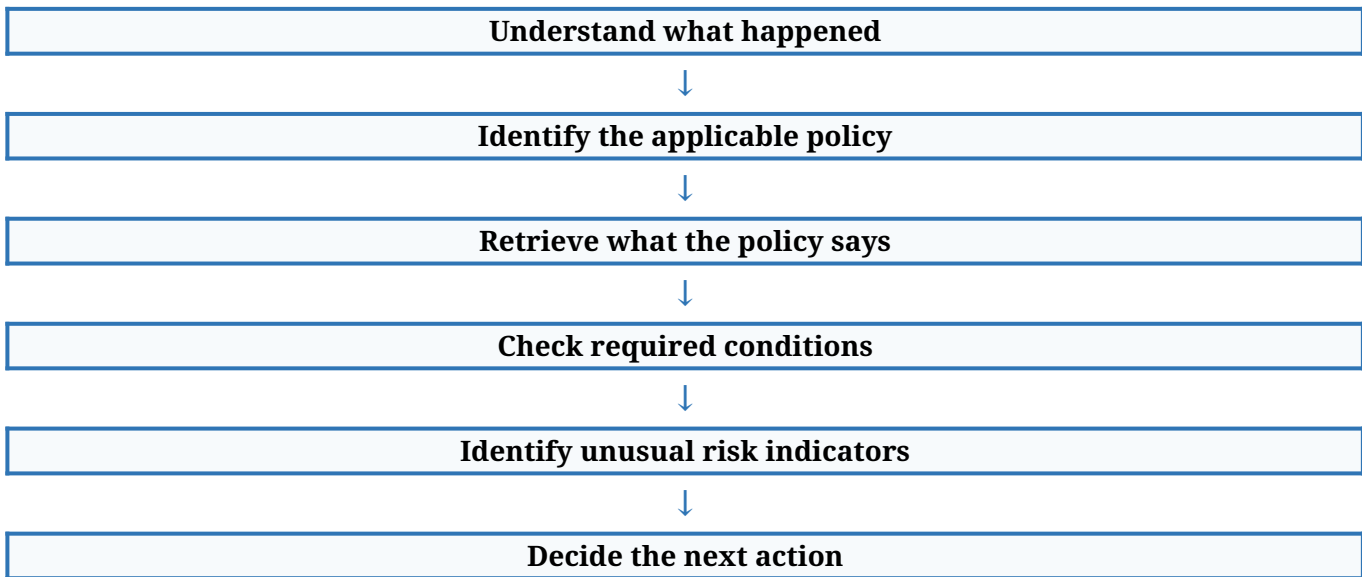
Motor insurance customers and claims teams work with lengthy policies, procedural rules, claim evidence and customer descriptions. Basic questions can still require manual searching and interpretation.

Customer challenges Policies are difficult to understand; customers may not know whether an incident is potentially covered, what documents are needed, or what steps to take.	Officer workload Claims officers often need to read free-text descriptions, search policy documents and organize claim details before meaningful review can begin.
Risk review Suspicious or inconsistent claims require additional checking and clear reasons for escalation.	Response delays Customers may wait for basic guidance even when the required information already exists in policy or claims documents.

Example customer question: “A stone hit my windscreen yesterday and created a small crack. Is this covered?”

A useful support system should help the customer identify the relevant policy section, understand possible coverage conditions, learn what documents are required and know the next step - without pretending to make the insurer’s final decision.

1.3 Why the Problem Matters



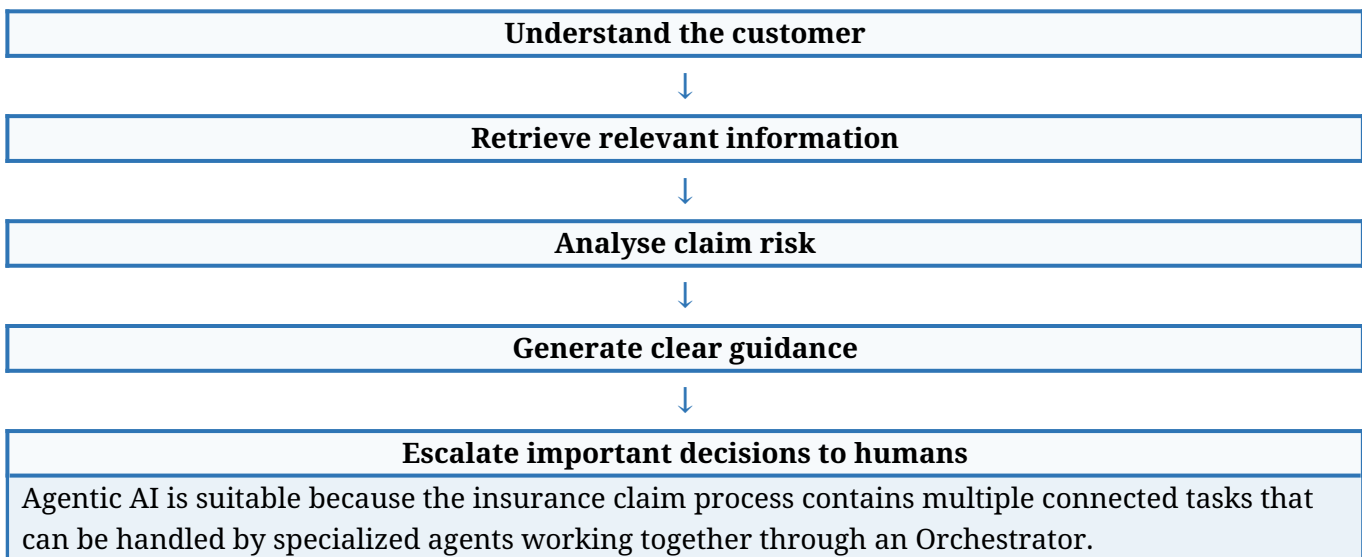
Doing all of these tasks manually can be slow. The proposed system assists with the early stages while retaining human control over final, consequential decisions.

1.4 Main Users

User	Primary Needs	System Support
Customer	Understandable policy information, claim guidance, document requirements, status updates	Natural-language interaction, evidence-based explanations, next-step guidance
Human Claims Officer	Structured claim data, relevant evidence, risk indicators, easier information access	Pre-organized claim information and explainable triage for human review

1.5 Why Agentic AI Is Suitable

The problem contains multiple connected tasks rather than one isolated prediction or chatbot response. Specialized agents can each perform a bounded responsibility while an Orchestrator coordinates the workflow.



2. Our Proposed System

Proposed system: An Agentic AI-based Motor Insurance Claims Processing and Policy Support System.

Customers interact using normal natural language. The system interprets the request, retrieves relevant insurance evidence, performs initial fraud-risk triage for actual claims, and provides appropriate guidance or escalation.

2.1 Example Interaction

“Someone hit my car yesterday near Kandy and damaged the left door. I want insurance to repair it.”

The system should infer that this is a motor claim request even though the customer may not use formal insurance terminology.

2.2 Target Users and Value

Customer Ask policy or coverage questions, understand claim procedures, submit initial claim information, receive status messages and guidance.	Human Claims Officer Review structured claim details, retrieved policy evidence and fraud-risk indicators; investigate when needed; make the final claim decision.
Customer value Faster and easier access to insurance information and early-stage claim guidance.	Insurer value Reduced repetitive manual work and better-organized information before human review.

3. Agents and Their Roles

System composition: 1 Orchestrator + 4 specialized AI Agents + Human Claims Officer.

3.1 Orchestrator - Central Workflow Controller

The Orchestrator manages where a request should go and what should happen next. It does not replace Agent 1's language-understanding role.

- Receive customer requests
- Manage authentication and authorization
- Validate and sanitize input
- Send requests to Agent 1 and receive structured results
- Select the correct workflow
- Route information between agents
- Track workflow state
- Escalate to a human claims officer
- Return results to the user interface

Important distinction: Agent 1 understands what the user wants. The Orchestrator decides what the system should do next.

3.2 Agent 1 - Claim Intake & Query Understanding Agent

Main technology: Natural Language Processing (NLP).

Agent 1 converts a customer's free-text message into structured information for downstream processing.

Example: "A bus hit my car yesterday and I want insurance to pay for the repairs."

Field	Possible Value
intent	claim_submission

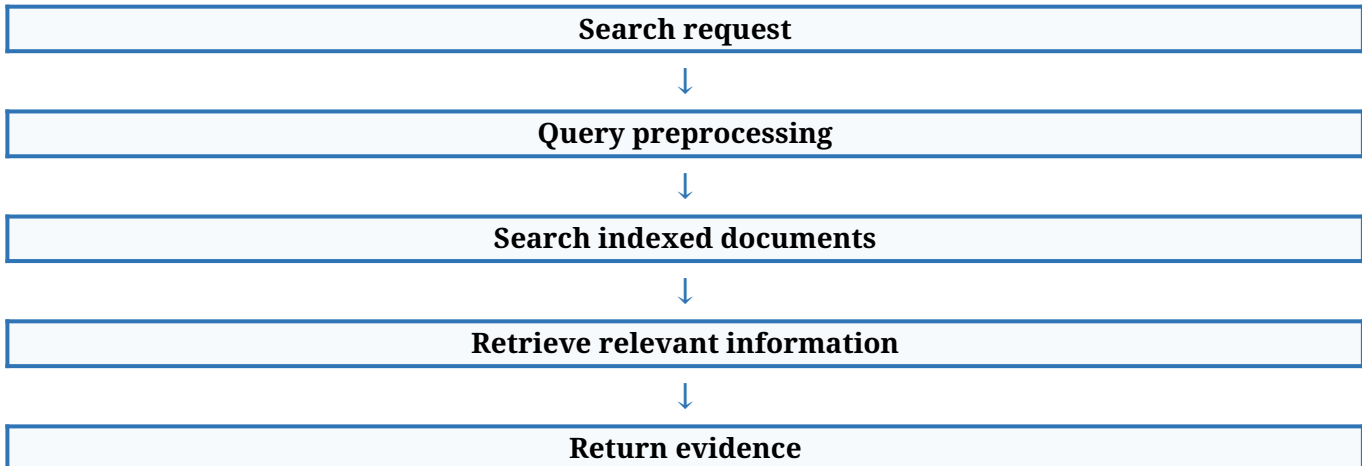
insurance_type	motor
incident	vehicle_collision
time	yesterday

It may also extract location, date/time, damage details and whether the user is asking for information or actually submitting a claim.

3.3 Agent 2 - Information Retrieval Agent

Agent 2 performs the Information Retrieval (IR) component. It searches a controlled insurance information collection for relevant evidence.

- Motor insurance policies
- Claim procedures
- Required documents
- Coverage conditions
- Policy exclusions
- Claim guidelines



Planned IR techniques include text preprocessing, inverted indexing, Boolean retrieval, positional indexing and tolerant retrieval where needed.

Key principle: Agent 2 finds insurance evidence. It should not use an LLM to invent policy facts.

3.4 Agent 3 - Fraud Triage Agent

Agent 3 is primarily used in the claim-submission workflow. Its purpose is to identify risk indicators or unusual patterns for human review - not to accuse a customer of fraud.

- Claim-date inconsistency
- Vehicle-detail mismatch
- Repeated or unusual claim patterns
- Information conflicts across submitted data
- Other controlled, relevant indicators

Example output: Risk level = Medium; reasons = accident-date inconsistency and vehicle-information mismatch; human review required = Yes.

Regardless of the risk level, the final claim decision remains with a human claims officer.

3.5 Agent 4 - Guidance Agent

Main technology: Large Language Model (LLM).

Agent 4 converts retrieved evidence into clear customer-facing explanations, next steps and status messages.

Core rule: IR retrieves first -> LLM explains second.

Status example: “Your claim has been received and is currently under review. A claims officer will contact you if additional information is required.”

3.6 Human Claims Officer

The Human Claims Officer is a formal part of the architecture. AI assists; the human retains authority over consequential decisions.

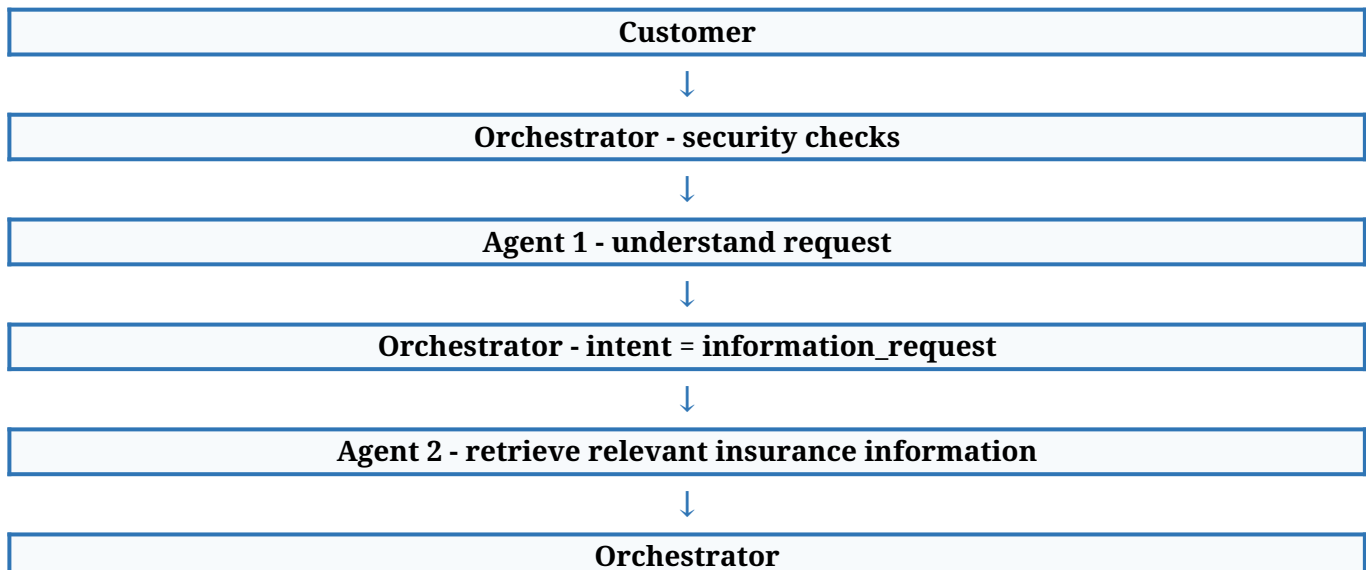
- Receives structured customer claim information
- Reviews relevant policy evidence and claim requirements
- Reviews fraud-risk indicators and their reasons
- Investigates where necessary
- Makes the final claim decision

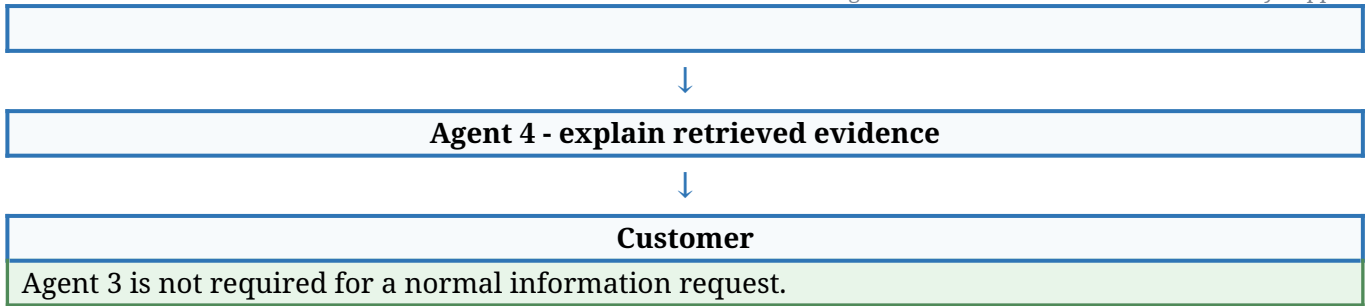
The system should not automatically approve or reject a claim, officially declare fraud, determine legal liability, or make the final compensation decision.

4. Complete Workflow

4.1 Information Request Workflow

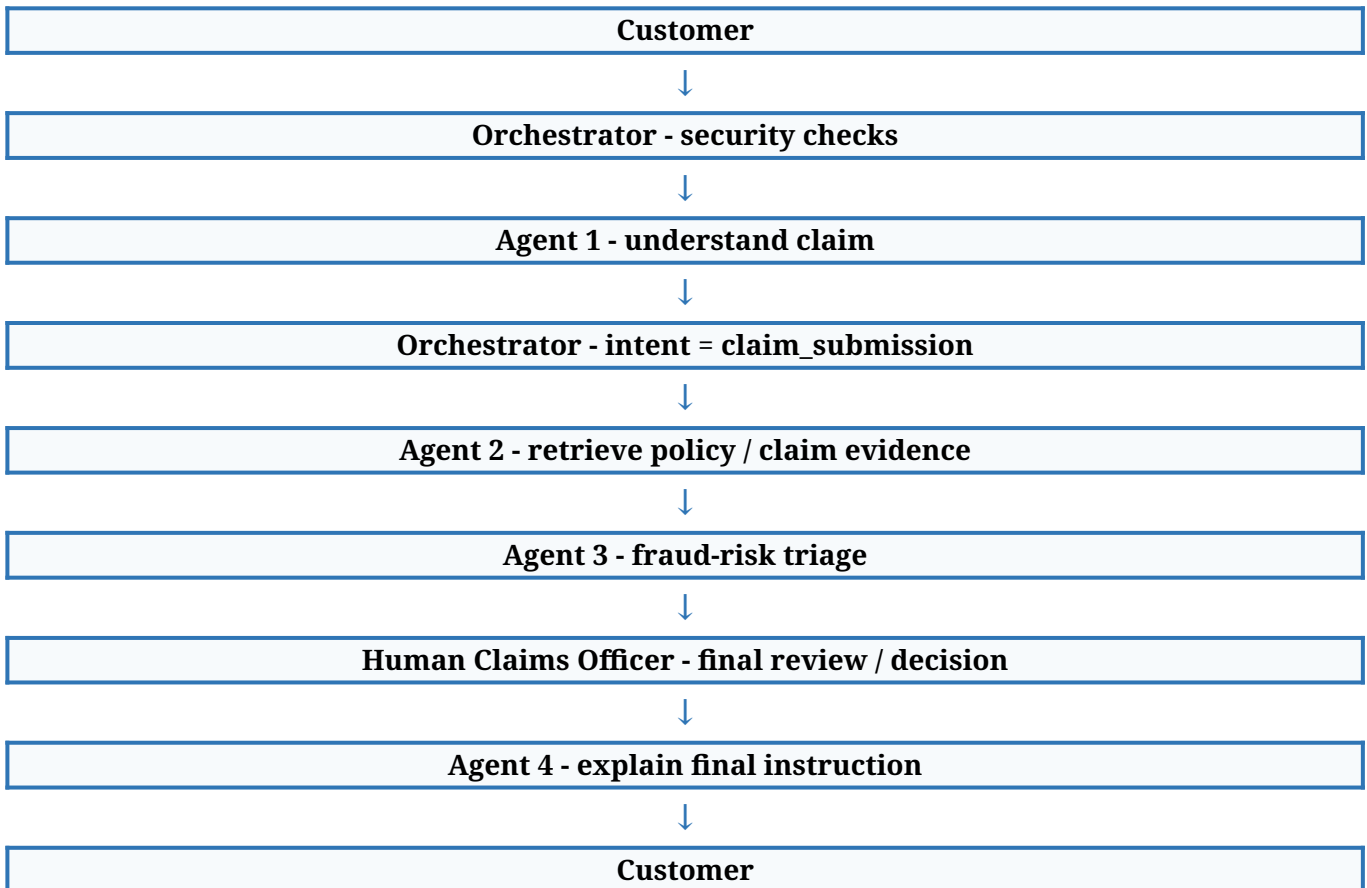
Example: “What documents do I need for a motor accident claim?”





4.2 Claim Request Workflow

Example: "Another vehicle hit me yesterday and I want insurance to repair my car."



In parallel with human review, Agent 4 may send a neutral claim-under-review status message. The architecture therefore supports timely communication without allowing the AI to imply that a final decision has already been made.

5. Implementation Plan

The Mid Evaluation does not require a complete production system. The project should clearly demonstrate how the proposed architecture could be implemented.

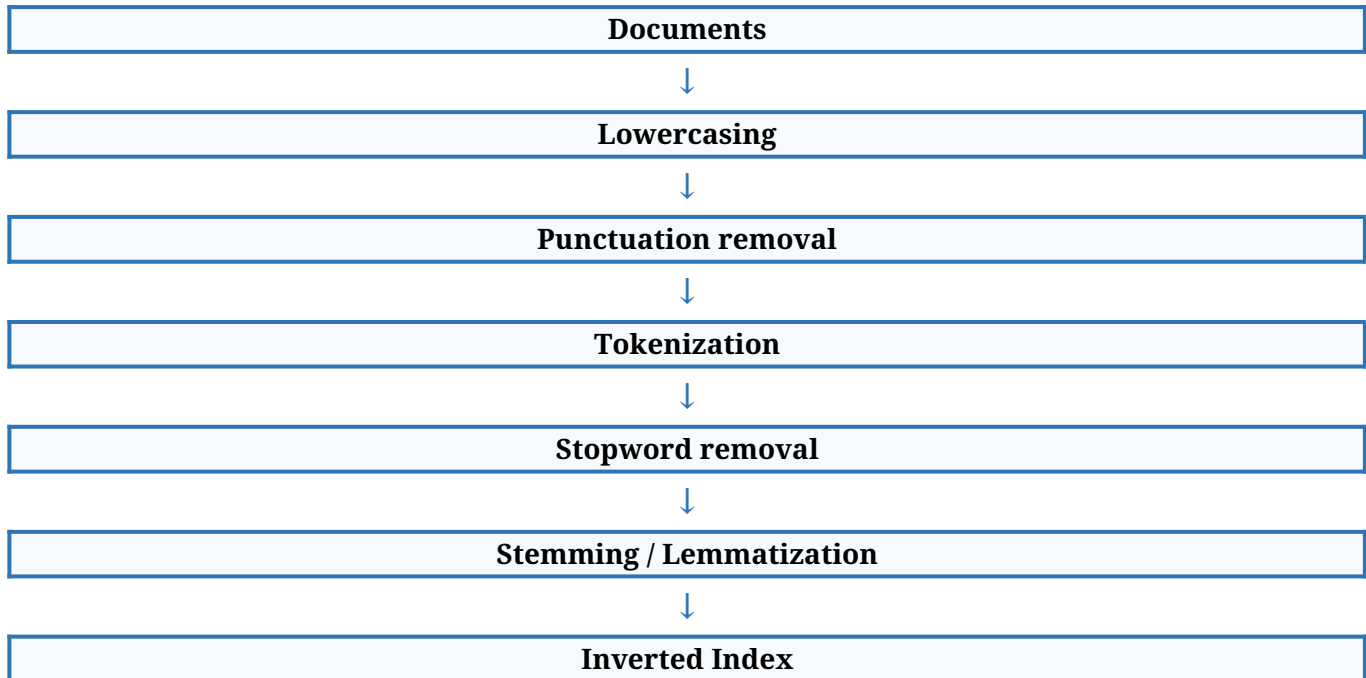
5.1 Build a Controlled Insurance Information Collection

- motor_policy.txt
- motor_claim_procedure.txt

- accident_claim_guide.txt
- windscreen_claim_guide.txt
- theft_claim_guide.txt
- required_documents.txt
- policy_exclusions.txt
- claims_faq.txt

The prototype can use publicly available insurance information and clearly labelled synthetic documents.

5.2 Build the Information Retrieval System



Suggested technologies: Python, NLTK, inverted indexing, positional indexing, Boolean retrieval and tolerant retrieval.

5.3 Develop Agent 1

Use Python and NLTK for preprocessing, then classify user requests into categories such as claim_submission, policy_question, coverage_question, required_documents_question and general_information. The exact classifier can be finalized during implementation.

5.4 Build the Orchestrator

Conceptual routing: if intent == “claim_submission”, run claim_flow(); elif intent == “information_request”, run information_flow().

The Orchestrator also manages security checks, state and communication between services.

5.5 Develop the Fraud Triage Agent

The first prototype can use controlled rules and available claim information. The output should include the indicator, risk level, reason and whether human review is required.

5.6 Connect the LLM to Agent 4

Agent 4 receives retrieved evidence and produces simple explanations, next-step guidance, status messages and other human-readable responses. The LLM is an explanation layer, not the source of policy truth.

5.7 Agent Communication

Recommended approach: REST API + JSON.

Field	Example
request_id	REQ001
intent	claim_submission
incident	vehicle_collision

5.8 Security Controls

- Authentication
- Authorization and role-based access control
- Input validation and sanitization
- Protected API keys and secure credential storage
- Secured customer and claim information
- Logging and controlled access to claims

Access example: Customer -> own claim only; Claims Officer -> assigned claims; Unauthorized user -> access denied.

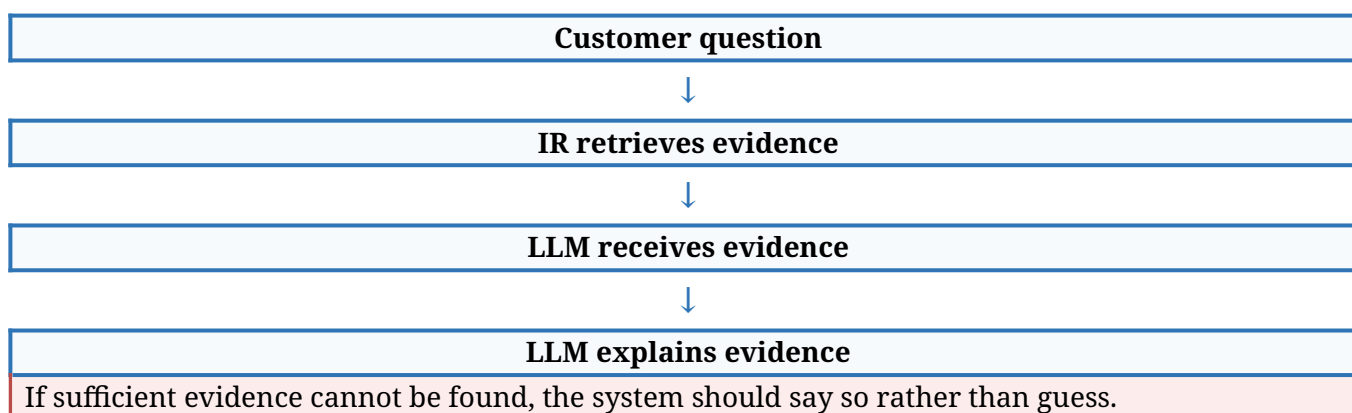
6. Responsible AI Plan

Responsible AI is especially important because insurance claims involve sensitive personal information and consequential decisions.

Principle	Application in This System
Fairness	Use relevant claim information and defined indicators. Avoid decisions based on irrelevant personal characteristics.
Transparency	Tell users that AI is assisting. Do not present AI guidance as a final human insurance decision.
Explainability	If Agent 3 flags risk, provide specific reasons so a human officer can understand the trigger.
Privacy	Restrict data access, authenticate users, minimize unnecessary personal data and prefer synthetic data for development/testing.
Security	Protect against unauthorized access, malicious

Principle	Application in This System
	input, data leakage and exposed API credentials.
LLM hallucination	Ground responses in retrieved evidence. If sufficient evidence is unavailable, state that clearly and escalate when necessary.
Fraud accusation risk	Use the language of risk indicators and human review rather than automatic accusations.
Human oversight	Keep final claim approval, rejection, liability and compensation decisions under human authority.
Responsible AI summary: AI = assistance; Human = authority.	

6.1 Hallucination Control Pattern



7. Commercialization Plan

7.1 Potential Customers

- Insurance companies
- Motor insurance providers
- Claims departments
- Customer-service departments
- Insurance brokers

Primary business model: B2B - Business to Business.

7.2 Value Proposition

- Reduce repetitive policy searches
- Reduce basic customer-service workload
- Provide faster initial claim support
- Organize claim information before review
- Help officers find relevant policy evidence

- Identify claims requiring closer review
- Provide customers with 24/7 initial guidance

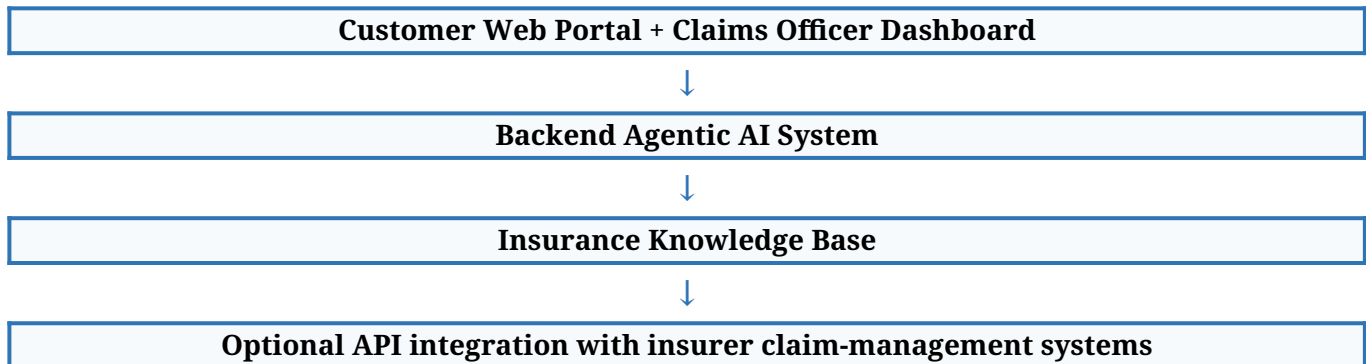
7.3 Revenue Model

Suggested model: Software-as-a-Service (SaaS) subscription.

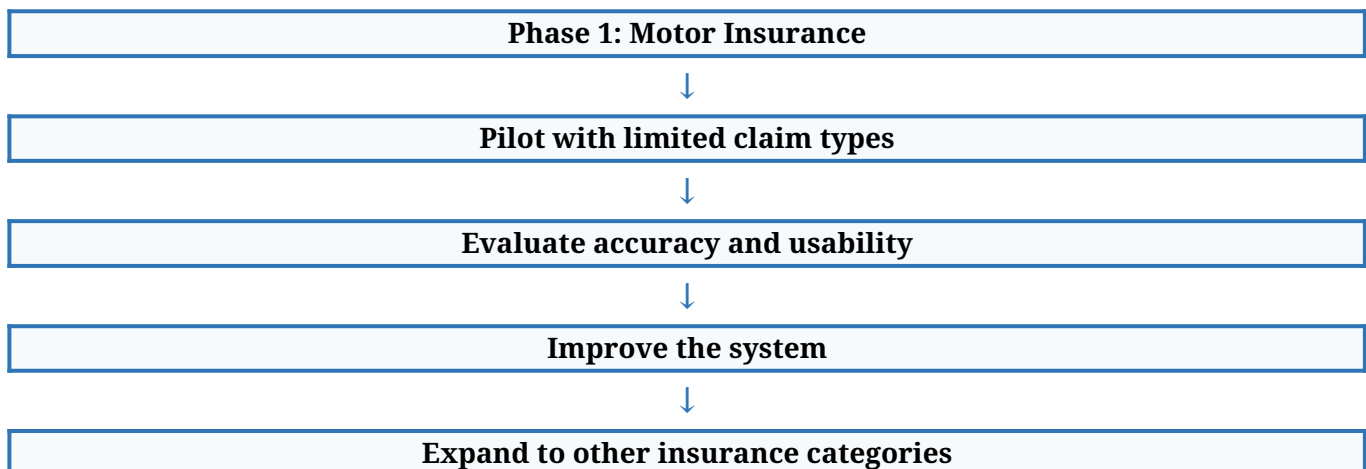
Pricing could scale by number of users, claims processed, usage volume or organization size.

Example tiers could be Starter, Professional and Enterprise.

7.4 Deployment



7.5 Market Introduction



A realistic prototype scope could cover vehicle collisions, windscreen damage, flood damage and theft/break-in cases.

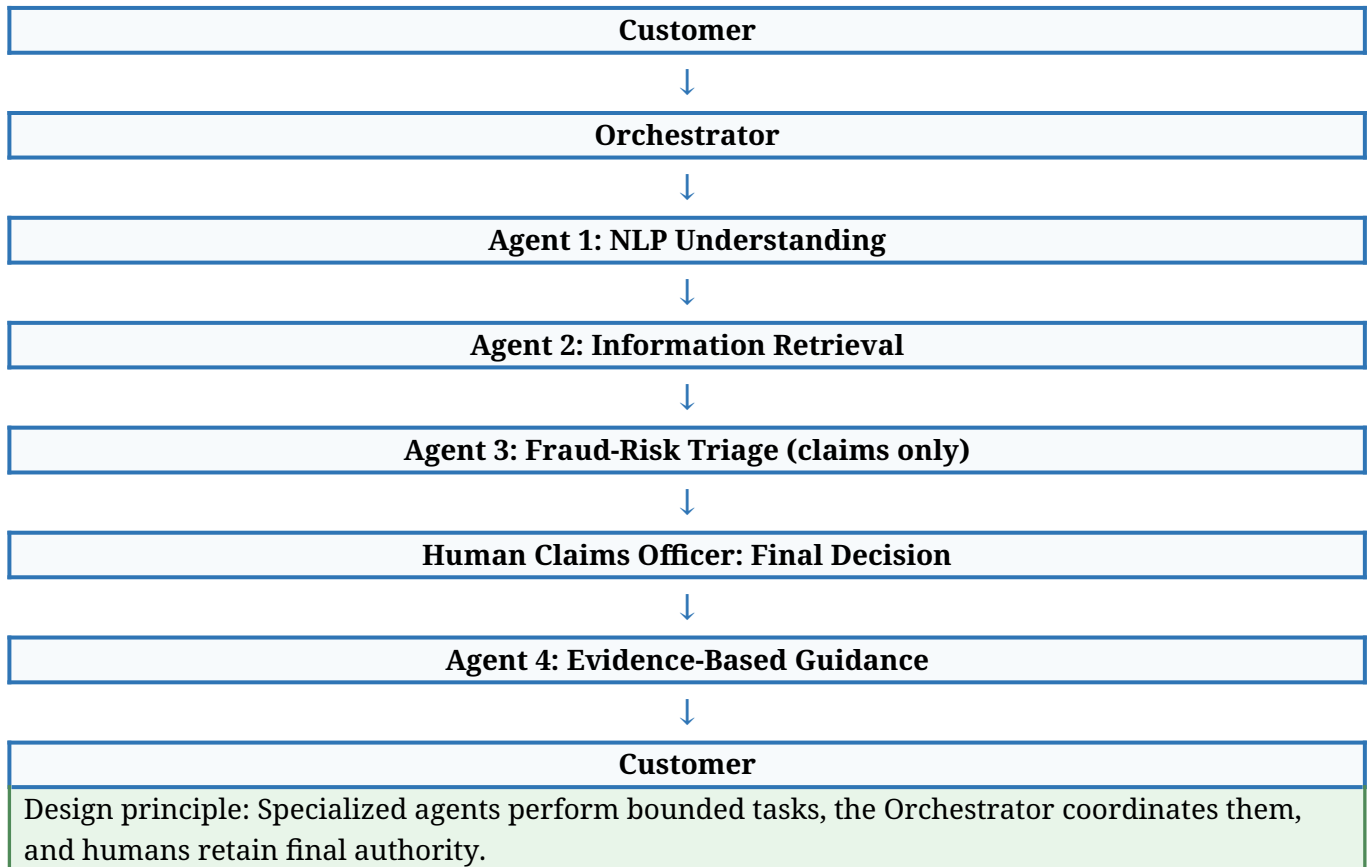
8. Complete Proposed Solution - Presentation-Ready Answer

Use the following as the core explanation every group member should understand.

We are proposing an Agentic AI Motor Insurance Claims Processing and Policy Support System. A customer's request first enters the Orchestrator, which manages security and workflow. The Orchestrator sends the request to Agent 1, which uses NLP to understand the customer's intent and extract important claim information. The result returns to the Orchestrator. If it is an information request, the Orchestrator sends it to the Information Retrieval Agent, which retrieves relevant policy or claim information, and the Guidance Agent uses an LLM to explain

that evidence to the customer. If it is an actual claim request, the IR Agent retrieves relevant policy and claim information, and the Fraud Triage Agent identifies potential risk indicators. The claim information, retrieved evidence and risk indicators are sent to a human claims officer, who has authority over the final claim decision. While the claim is being reviewed, the Guidance Agent sends the customer a neutral status update. This architecture allows us to combine NLP, information retrieval, LLMs, security and multi-agent communication while keeping important insurance decisions under human control.

8.1 Architecture at a Glance



End of Proposed System Document